Non-Linear Tube Saturation & Overdrive DSP

Mathematical Models of `[drive~]` and Waveshaping Objects

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Abstract

This paper formulates the non-linear transfer functions, hyperbolic tangent $\tanh(x)$ waveshaping, asymmetric grid bias tube saturation equations, cubic soft-clipping polynomials, and harmonic gain normalization used in the `[drive~]` node object.

1 Symmetric Hyperbolic Tangent ($\tanh$) Saturation

In vacuum tube triode amplification, soft saturation limits signal amplitude as input voltage increases. The symmetric transfer function $f(x)$ for input signal $x(n)$ with drive gain $G \geq 1.0$ is:

$$y(n) = \tanh\left(G \cdot x(n)\right) \tag{1}$$

Taking the Taylor series expansion of $\tanh(u)$ around $u = 0$:

$$\tanh(u) = u - \frac{u^3}{3} + \frac{2u^5}{15} - \frac{17u^7}{315} + O(u^9) \tag{2}$$

The presence of odd-powered polynomial terms (u^3, u^5, u^7) introduces odd harmonic overtones $(3f, 5f, 7f)$ into the output spectrum, generating warm analog saturation.

2 Asymmetric Triode Vacuum Tube Bias Modeling

To simulate authentic vacuum tube asymmetric clipping (which introduces even harmonics $2f, 4f, 6f$ associated with tubey warmth), a DC grid bias offset $V_{\text{bias}} \in [-0.5, 0.5]$ is injected prior to non-linear transformation:

$$y_{\text{raw}}(n) = \tanh\left(G \cdot x(n) + V_{\text{bias}}\right) - \tanh(V_{\text{bias}}) \tag{3}$$

Subtracting $\tanh(V_{\text{bias}})$ removes the static DC offset, preserving zero DC operating bias at rest.

3 Cubic Soft-Clipping Polynomial

For lightweight processing without transcendental functions:

$$y_{\text{cubic}}(x) = \begin{cases} -1.0, & x \leq -1.5 \\ x - \frac{x^3}{3}, & -1.5 < x < 1.5 \\ 1.0, & x \geq 1.5 \end{cases} \tag{4}$$

4 Harmonic Gain Compensation & Peak Normalization

To maintain consistent output loudness across drive settings $G \in [1.0, 100.0]$, an dynamic automatic gain compensation multiplier $A(G)$ is applied:

$$A(G) = \frac{1}{\max(1.0,\ \tanh(G))} \approx \frac{1.0}{\sqrt{G}} \tag{5}$$

The final normalized output $y_{\text{final}}(n)$ is:

$$y_{\text{final}}(n) = A(G) \cdot y_{\text{raw}}(n) \tag{6}$$